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Atmosphere

I Keywords

1	Troposphere	6.	Density
2.	Stratosphere	7.	Green house
3.	Mesosphere	8.	Ultra violet
4.	Thermosphere	9.	Ozone layer
5.	Exosphere	10.	Heat waves
		11.	Biodiversity.

II Terms to remember

1. Density: mass per unit

2. ultra violet rays: invisible rays which are a part of solar radiation and can harm living organisms on earth.

3. heat wave: a very long period of hot weather

4. biodiversity: rage of diffrent forms of plant and animal life in a region.

III Teaching Points

1 Atmosphere : the blanket of air which surrounds the earth.
- It is held due to gravity
- Density of air decrases with imcreas in height

2 Composition of atmosphere.

- Mixture of colourless, odourless and tasteless gases.

- Composed of gasses, water vapour, dust particles and smoke.
- The gasses include nitrogen, argon, oxygen, carbon dioxide, neon, ozone, hydrogen, helium.

3 Other facts of atmosphere.

- Heavier gases like oxygen, water vapour, and dust particles are found in lower layers of atmosphere.

- At many places impurities like smoke, salts, carbon monoxide and other chemicals are also found.

Layers of atmosphere.

- Troposphere
- Stratosphere
- Mesosphere
- Thermo's phere
- Exosphere

IV Concept map

Structure of Atmosphere.

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Troposphere
Height : 8 km - 18 km
Temp : Decreses
Water : Vapour dust particles is found.
All weather phenomenon rain, thunder, etc.

Stratosphere
Height 50 km , ozone layer
Temp : Increases
No weather phenomenon
jet planes fly here
Movement air - of horizontal.

Mesosphere
Height : 50 km - 80 km
Temp : Decrease (-100°c)
Coldest layer
Meteors burn up

Thermosphere
Height 80 km - 400 km
Temp : Increases upto 2000°c
Contains : Ionized molecules
which reflect radio waves
and helps in long distance
communication [Ionsphere]

↓

Exosphere

Height : Merges into the Space
Temp : Increases upto 4500°C
Lighter gases. Hydrogen, Helium.

Importance of atmosphere.

* The lower layer of the atmosphere has oxygen which is essential for breathing.

* It helps in the change of weather.

* It helps to transmit sound waves.

* It sheilds us from meteors.

* It protects us from harmful ultra violet rays of the sun.

Greenhouse effects

* A part of heat is trapped in the atmosphere by gases such as carbon-dioxide, methane, nitrous oxide, water vapour, chlorofluorances, while the rest escapes into space, these gases are called greenhouse gases.

* A french scientist J.B.J Fauries assumed the earth as a giant greenhouse.

* The earth's atmosphere traps the heat energy that comes from the sun and prevents the earth from becoming too cold at night is known as the greenhouse effect.

Global warming

* The excess accumulation of green house gasses in the atmosphere, disturbs the energy balance which can increase the heat and cause rise in air temperature is called global Warming.

* The four major activities that produce greenhouse gases are agriculture, energy use, industrial process and deforestation

Impact of global warming

Rise in sea level

Change in agriculture

Climate change

Shortage of freshwater

Increased health risk

Loss of biodiversity

Increase in evaporation rate.

Measures taken to control global warming

* Reduction in quality of chemical fertilizers and insecticides used to grow crops.

* Use of alternative energy resources instead of coal and petroleum.

* Redultion in Deforestation

* Planting more trees

* Finding alternative materials (Hydrofluro carbons to reduce release of chlorofluro carbans)

HW - 1

1 Global warming is a matter of concern today. Explain in detail

Ans * Rise in sea level - It would speed up melting of glaciers. This will lead to rise in sea level causing floods.

* Climate change : As the earth heats up, the world climatic patterns will change. The frequency of floods and storms could threaten coastal communities, while heat waves, fires and droughts also become more common.

* Shortage of fresh water : Freshwater resources may be deminished due to shrinking mountains etc. Rising sea level may pollute freshwater with saltwater.

* Loss of Biodiversity - Most plant and animals grow and develope in a specific temp range. The global warming may force many of them to shift slowly towards high latitude. Many of the species became extinct.

OZONE

* A layer in the earth's atmosphere at an altitude of about 10 km (6.2 miles) containing a high concentration of ozone, which absorbs most of the ultraviolet radiation reaching the earth from the sun.

* Average thickness - 2.9 milli metres

* If the thickness of ozone layer in the stratosphere falls below 220 DU then it is called as ozone hole.

What is DOBSIN unit ?

Ans. • Unit of measurement used to measure the thickness of ozone layer.

• The normal thickness of ozone layer is 2.9 mm will be equal to 220 DU

• One dobsin unit is defined to be defined 0.01 mm

Deplition in ozone layer

* Can allow more ultraviolet Rays to reach the earth's surface.

* The negative influences due to depletion of ozone layer are:

1) Skin cancer
2) Immune deficiency
3) Respiratory illness
4) Disturbed plant life cycle
5) Disturbed marine life

* The main pollutants responsible for the deplition of the ozone layer are:

- Chlorofluro carbons
- Nitrogen oxide
- Hydro carbons.

Prevention of Ozone Hole

* In September 1987, 34 industrialized nations signed the Montreal Protocol.

* old refrigerators and air conditions are disposed safely.

* The ozone depleting substances should'nt release in the atmosphere.

* When repairing or renovating our houses, make sure that old insulation forms containing ozone depleting substances are disposed of as environmentally hazardous waste.

Qs. What is ozone hole? What was the aim of Montreal protocol?

Ans If the thickness of the ozone layer in the stratosphere fells below 220 DU, then it is termed as ozone hole.
The montreal protocol was aimed to phase out use of ozone-depleting substances and to find non-toxic substances.

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Q1 State three functions of atmosphere?

Ans • It protects us from harmful UV rays.
• It keeps earth surface warm even at night.
• It softens Sun's glare during the day.

Q2. Suggest any 3 practical steps that you and your family can take at home to reduce ozone layer depletion.

Ans • To make sure that old refrijirator and AC's are safely disposed, by giving them to recycle yard.

• Repairing or renovating your house, old insulation foams containing ozone depleting substances are disposed of enviroment hazardous waste.